

Shared Hub · Scores at a Glance

Zero-shot head transfer across encoders, and held-out encoders added to the frozen hub

How to read this. One caption-retrieval head is trained on ONE encoder's hub coordinates (DINOv2-small), then applied UNCHANGED to encoders it has never seen. 'Percent of native' is that frozen head's R@1 as a fraction of a head fitted specifically for that encoder. 100% = the frozen head loses nothing. A random map replaces the fitted one as a control and sits at chance (0.001). The hub itself is never rebuilt.

1 · Hub members · the spaces inside the concatenation

The published hub is 4 spaces (three DINOv2 + bge-m3); the 7-space variant adds GPT-2, BERT and SBERT. Image rows are img->txt retrieval; text rows are the read-out targets and sources.

encoder	width	raw R@1	% of native	role / note
DINOv2-small	768	0.398	111.5%	head trained here (>100% = its native ref is weakest)
DINOv2-base	1536	0.442	95.9%	zero-shot, never seen by the head
DINOv2-large	2048	0.468	92.9%	zero-shot; conservative (entry map 4.2 rows/dim)
bge-m3	1024	.	.	the read-out TARGET; anisotropic (+0.57) but writable 98.1%
SBERT	768	.	101.0%	cleanest text frame (+0.135); marginally better target
BERT	768	.	76.6%	writable, anisotropic; masked-LM
GPT-2	768	0.095	27%	readable 95% but collapses on transfer (rank 6.6)
random-map control	.	0.000	chance	0.001 = 1/1000; proves the protocol is honest

Source: Report Table C12 (transfer), Table C12f (target choice), C.11/C.14 (GPT-2 readable-not-writable).

Readable vs writable · why GPT-2 is in the hub yet is a poor target. Readable = a linear map can pull content OUT of a space (GPT-2: 95%). Writable = cosine can rank items once you predict back INTO it (GPT-2: 25%). Reading needs recoverable content (effective rank); writing needs items cosine can separate (pair-cosine). A target must be writable, which is why the target comparison uses writability, and why bge/SBERT qualify and GPT-2 cannot.

Does the hub need the text side? G10: a hub built images-only transfers at 85.5% of native; adding bge-m3 lifts it to 95.3% · a 9.8-point gain. The shared space is not an image-only artifact.

2 · Held-out encoders · never in the hub

Each is encoded over the same items, given ONE linear map into the FROZEN hub, and read by the SAME frozen head. Nothing upstream is refitted. This is the extensibility test: can the hub absorb an encoder of unrelated lineage or architecture at one closed-form solve?

Two protocols, two columns. 'Published' is the original G-series head; 'rebuilt' is the reconstructed head after the VM loss. They differ by a measured offset (ConvNeXt reads +4.5 pts under the scale-extension protocol), so compare within a column, not across. Both are shown for honesty.

held-out encoder	published	rebuilt	what it establishes
SigLIP 2	94.2%	86.4%	ViT, fully separate lineage · shared DINOv2 ancestry was NOT what made transfer work. Also the calibration control.
ConvNeXt-base	96.7%	87.5%	CONVOLUTIONAL · no attention, no patches. Architecture does not bound the claim. (101.2% under scale protocol = noise, McNemar p=0.668.)
DINOv2-giant	100.0%	·	1.14B, cls+patch. INCONCLUSIVE: entry map at 2.8 rows/dim is underpowered; ratio hits 1.0 by degeneracy, not excellence.
Qwen3-Emb-0.6B	93.7%	·	text->text task (not comparable to img->txt). Architecture control, 1.06x bge-m3.
Qwen3-Emb-4B	96.2%	·	text->text, 4B. INCONCLUSIVE: 3.3 rows/dim underpowered.

Sources: Report §E.2 & G4 (SigLIP, ConvNeXt), scale-extension run (giant, Qwen). Within-family band 93.8-96.5%.

3 · Scale extension · does alignment strengthen with capacity?

Pre-registered, both modalities, matched pooling. Raw shape agreement (rho), no map fitted. Both ladders rise then FALL:

ladder	peak	largest model	verdict
DINOv2 image (0.02-1.14B)	0.418 (large)	0.339 (giant)	P2 FALSIFIED
contrastive text (0.57-4B)	0.352 (Qwen-0.6B)	0.307 (Qwen-4B)	P3 FALSIFIED

Reading. Cross-modal alignment does not increase monotonically with capacity in the 0.02-4B regime; it peaks then declines on BOTH modalities. Within-modality, contrastive text encoders agree with each other at 0.76-0.86 (the highest pairs in the matrix) while their cross-modal agreement stays 0.19-0.35. This is a measured answer to the scale caveat, not a concession

- Huh et al.'s 70B regime is a different one.